

Task_Forms_Phase2

Phase 2 — Weekly Task Forms (one form per member)

Status reflects the repository on 2026-09-04. "Completed" means code, run logs and results exist in the repo; see [docs/Phase2_Report.md](#) for the numbers behind each row.

Form 1 — Vishal Kumar (2023011085)

S.No.	Week	Task Assigned	Task Status
1	Week 1	Study Federated Learning concept and FedAvg algorithm	Completed
2	Week 2	Set up Flower (flwr 1.36) framework in gRPC deployment mode; server + 4 client processes	Completed
3	Week 3	Design overall federated system architecture (<code>federated/config.py</code> , <code>server/client/orchestrator</code>)	Completed
4	Week 4	Implement TLS between nodes and server (self-signed CA, server certificate, client verification)	Completed
5	Week 6	Integrate hybrid model (XGBoost + Autoencoder) into the federated pipeline; redesign decision rule	Completed
6	Week 8	Compare centralized vs federated accuracy and performance (<code>federated/evaluate.py</code>)	Completed
7	Week 10	Final system testing (16 tests), Phase 1 audit fixes, project documentation	Completed

Form 2 — Shashwat Srivastava (2023021159)

S.No.	Week	Task Assigned	Task Status
1	Week 1	Study CIC-IDS2017 dataset structure and attack types (all 8 capture days)	Completed
2	Week 2	Load and preprocess multiple attack datasets (DDoS, DoS, PortScan, BruteForce, Bot, WebAttack)	Completed
3	Week 3	Distribute dataset across 4 simulated federated nodes with calibration/report hold-out	Completed
4	Week 4	Handle data heterogeneity (non-IID nodes by attack family, per-node class weighting)	Completed
5	Week 6	Integrate WebAttack and Infiltration data; CSE-CIC-IDS2018 schema mapping	Completed (2018 mapping untested: files not available)
6	Week 8	Evaluate detection accuracy per attack family on multi-attack test data	Completed

S.No.	Week	Task Assigned	Task Status
7	Week 10	Dataset documentation and report section (Chapter 4)	Completed

Form 3 — Tarkeshvar Mani Yadav (2023011078)

S.No.	Week	Task Assigned	Task Status
1	Week 1	Study local model training in a federated setup	Completed
2	Week 2	Implement local XGBoost training on each node (bagging client)	Completed
3	Week 3	Implement local Autoencoder training on each node (FedAvg client)	Completed
4	Week 5	Add differential privacy to model aggregation (fixed clipping + Gaussian noise)	Completed; result negative with 4 nodes (Report §7.4)
5	Week 6	Test local model performance before and after aggregation (per-round client and server metrics)	Completed
6	Week 8	Measure communication overhead per federated round (<code>federated/comm_analysis.py</code>)	Completed
7	Week 10	Algorithm section of the report (Chapters 5–6), presentation	Completed

Form 4 — Ritesh Pandey (2023021154)

S.No.	Week	Task Assigned	Task Status
1	Week 1	Study FedAvg and secure aggregation techniques	Completed
2	Week 2	Implement FedAvg global aggregation on the central server; XGBoost bagging aggregation	Completed
3	Week 3	Set up 4 simulated federated client nodes locally (<code>federated/run_federated.py</code>)	Completed
4	Week 5	Test model convergence across federated rounds (per-round JSON logs, convergence analysis)	Completed
5	Week 6	Secure model-update transmission (TLS channel; server-side threshold calibration)	Completed
6	Week 8	Update IDS dashboard: federated node status, round curves, DP/LSTM tables, prevention panel	Completed
7	Week 10	Final integration testing (Streamlit AppTest), submission preparation	Completed

Items beyond the 10-week forms (from the Phase 1 report's future-work list)

Item	Status
Autonomous prevention (IP blocking, firewall rules)	Completed: <code>core/prevention.py</code> , dry-run default, audit + undo
LIME alongside SHAP	Completed: LIME-style local surrogate of the hybrid score
Persistent logging	Completed: SQLite event store
LSTM temporal model	Experiment completed; negative result, not adopted (Report §7.5)
CICIDS2018 support	Schema mapping implemented; not run (dataset not on disk)
Blockchain-verified model updates	Not attempted (out of scope for Phase 2)